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13 May.

• Writing a number in $\frac{P}{Q}$ form.

Case -I- when a no. have terminating decimal expansion.

eg - • 1.234

$= \frac{1234}{1000}$ > Integers

• 3.12789

$= \frac{312789}{100000}$ > Integers

Note → $\frac{p}{q}$ → p and q → co-prime

Eg → $2.5 = \frac{25}{10} = \frac{5}{2}$ Ans

Case - II

· when numbers have non-terminating and repeating decimal expansions.

1st type →

Eg → $1.343434.. \Rightarrow 1.\overline{34}$, $1.\overline{237}$, $23.\overline{6}$

2nd type →

Eg - $1.3\overline{4} \rightarrow 1.344...$

$1.2\overline{37} \rightarrow 1.2373737...$

$23.\overline{6} \rightarrow 23.666.....$

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Eg-7.

$0.33\ldots = 0.\overline{3}$ → Express in the form of $\frac{p}{q}$.

sol: ~~0.333~~...

$\Rightarrow x = 0.\overline{3}..$

$x = 0.333\ldots$ —(1)

• Multiply by 10, Eq^n (1)

$\Rightarrow 10x = 3.333\ldots$ —(2)

& (2 − 1)

$10x = 3.\cancel{333}\ldots$

$x = 0.\cancel{333}\ldots$

$9x = 3$

$\boxed{x = \frac{3}{9} = \frac{1}{3}}$

steps

1.) Equate the question to any variable (x) and mark it as Eq^n —(1)

2.) Multiply the Eq^n by power of 10 with respect

3.) the no. of digit on which bar is visible.

3.) subtract Eq(1) from Eq(2)

Eg-8

$1.\overline{27} \rightarrow \frac{p}{q}$ ⇒ Express it in the form of $\frac{p}{q}$

$x = 1.\overline{27}$

$x = 1.272727\ldots$ —(1)

• Multiply by 100, (Eq −(1))

$100x = 127.2727$ —(2)

• substract ~~eq~~ eq 1 from eq 2

$$\left[\begin{array}{l} 100x = 127.2727\ldots \\ x = 1.2727 \end{array}\right] -$$

$99x = 126$

$x = \frac{\cancel{126}\,42}{\cancel{99}\,33} = \frac{\cancel{42}\,14}{\cancel{33}\,11} \boxed{= \frac{14}{11}}$

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$$\begin{array}{r} 432.3232 \\ 4.32 \\ \hline 428 \end{array}$$

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Q1.) $4.\overline{32} \rightarrow$ in $\frac{p}{q}$ form

sol:

$x = 4.\overline{32}$

$x = 4.32\,32\,32\ldots$ — ①

Multiply by 100, Eq ①

$100x = 432.3232\ldots$ — ②

subtract eq ① from eq ②

$100x - x$

$432.32\ldots - 4.32\ldots$

$99x \Rightarrow 428$

$$\boxed{x = \frac{428}{99}}$$

Q2.) $2.\overline{317}$

sol:

$x = 2.\overline{317}$

$x = 2.317317\ldots$ — ①

• Multiply by 1000, Eq ①

$1000x = 2317.317\,317\ldots$

subtract eq ① from eq ②

$1000x = 2317.317\,317\ldots$

$x = 2.317\ldots$

2315.00

$999x = 2315$

$$\boxed{x = \frac{2315}{999}}$$

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Q3.) $0.666\ldots$ in the form of $\frac{p}{q}$ is $A \rightarrow \frac{6}{99}$, $B \rightarrow \frac{2}{3}$ ✓

$C \rightarrow \frac{3}{5}$, $D \rightarrow \frac{1}{66}$

sol: $x = 0.\overline{6}$

$x = 0.666\ldots$ — ①

• Multiply by 10, Eq ①

$10x = 6.66\ldots$ — ②

• subtract eq ① from eq ②

$10x = 6.66\ldots$

$x = 0.66\ldots$

$9x = 6$

$x = \frac{\cancel{6}\,2}{\cancel{9}\,3}$

$$\boxed{x = \frac{2}{3}}$$

Q4.

sol: $1.\overline{32}$

$x = 1.\overline{32}$

$x = 1.3232\ldots$ — ①

• Multiply by 100, Eq - 1

$100x = 132.32\ldots$ — ②

• subtract eq 1 from eq ②

$100x = 132.32\ldots$

$x = 1.32\ldots$

$99x = 131$

$$\boxed{x = \frac{131}{99}}$$

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1.97979797
0.34343434
2.32323232

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• $0.\overline{35}$

$y = 0.\overline{35}$

$y = 0.3535\ldots$ —①

• Multiply eq ① by 100

$100y = 35.35\ldots$ —②

• subtract eq ① from eq ②

$100y = 35.35\ldots$

$y = 0.35\ldots$

$99y = 35.00$

$\boxed{y = \frac{35}{99}}$

• Adding ① and ②

$x + y$

$\frac{131}{99} + \frac{35}{99} = \frac{166}{99}$

Homework

Ex-1.3

Q2.

a. $0.\overline{6}$

sol: $x = 0.\overline{6}$

$x = 0.666\ldots$ —①

• Multiply ~~10~~ by 10, ~~Eq^n~~ Eq^n —①

$10x = 6.6\ldots$ —②

• subtract eq ① from eq ②

$10x = 6.66\ldots$

$x = 0.66\ldots$

$9x = 6$

$\boxed{x = \frac{\cancel{6}2}{\cancel{9}3} = \frac{2}{3}}$

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b. $0.4\overline{7}$

$x = 0.4\overline{7}$

$x = 0.4777\ldots$ —①

• Multiply 10 in both sides in Eq ①

$10x = 4.77\ldots$ —②

• subtract eq 1 from eq 2

$10x = 4.777\ldots$

$x = 0.477\ldots$

$9x = 4.3$

$\boxed{x = \frac{43}{90}}$

c. $0.\overline{001}$

$x = 0.\overline{001}$

$x = 0.001001001\ldots$ —①

• Multiply by 1000, Eq ①

$1000x = 1.001001\ldots$ —②

• subtract eq 1 from eq 2.

$1000x = 1.001001\ldots$

$x = 0.001001\ldots$

$999x = 1$

$\boxed{x = \frac{1}{999}}$

Q4.) Express 0.999... in $\frac{p}{q}$ form

sol: —

$x = 0.\overline{9}$

$x = 0.999...$ —①

• Multiply by 10, Eq ①

$10x = 9.999...$ —②

• subtract eq 1 from eq 2

$10x = 9.99...$

$x = 0.99...$

$9x = 9$

$\boxed{x = \frac{9}{9} = 1}$

EX-1.1

Q1.

sol: Yes, we can write 0 in the form of $\frac{p}{q}$

$\Rightarrow \frac{0}{1} = 0 \quad \left[\frac{0}{1} = \frac{p}{q}\right]$

Q4. True / False

sol:

a.) Every natural number is a whole no. → True

b.) Every integer is a whole no. → No

c.) Every rational no. is a whole no. → No
(fraction)

Reasons

1.) As natural no. starts from 1 to ∞, and whole no. starts from 0 to ∞ so it contains all the natural no.

2.) Integers → include negative no. also but whole no. doesn't.

3.) rational no. includes fractions but not all fractions are whole no like →

• $\left[4 \Rightarrow \frac{4}{1}\right]$ → whole as well as rational no.

• $\left[\frac{5}{2}\right]$ → only rational no.

short trick

Eg-1 $\rightarrow 1.\overline{23}$

$\Rightarrow \frac{123-1}{99} = \frac{122}{99}$

Eg-2 $\rightarrow 23.\overline{14}$

$\Rightarrow \frac{2314-23}{99} = \frac{2291}{99}$

Eg-3 $\rightarrow 314.\overline{213}$

$\Rightarrow \frac{314213-314}{999} = \frac{313899}{999}$

Eg-4 $\rightarrow 1.\overline{336}$

$\Rightarrow \frac{1336-1}{999} = \frac{1335}{999}$

2nd type

Q. Express $1.2\overline{3}$ in $\frac{p}{q}$ form

sol: $x = 1.2\overline{3} \rightarrow$ bar on 1 digit

$x = 1.233\ldots$ —①

• Multiply by 10, Eq ①

$10x = 12.33\ldots$ —②

② - ①

$10x = 12.33\ldots$

$x = 1.23\ldots$

$9x = 11.1$

$x = \frac{111}{90}$

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EX-1.3

Q3.

2.) $0.4\overline{7}$ in $\frac{p}{q}$ form

$x = 0.4\overline{7}$

$x = 0.477\ldots$ —①

Multiply by 10, Eq ①

$10x = 4.77\ldots$ —②

② - ①

$10x = 4.77\ldots$

$x = 0.47\ldots$

$9x = 4.3$

$\boxed{x = \frac{43}{90}}$

Q. $13.1\overline{45}$

$x = 13.1\overline{45}$

$x = 13.145454545\ldots$ —①

• Multiply by 100, Eq-1

$100x = 1314.54545\ldots$ —②

subtract eq 1 from eq 2

$100x = 1314.54545\ldots$

$x = 13.145454545\ldots$

$99x = 1301.4$

$x = \frac{13014}{990} = \frac{13014}{990}$

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